

ClearLedger

System/Prototype Description · Bhoomika Ravishankar · IS455 – Database Design & Prototyping

1. System Name

ClearLedger is a personal finance tracking system. The name reflects its purpose: giving users a clear, organized view of their financial activity in one place.

2. Users

ClearLedger is built for individuals who want more structure in managing their money, students tracking limited income, young professionals starting to budget, or anyone tired of juggling spreadsheets and bank apps. Each user manages their own data privately; there is no shared or multi-user access in this prototype.

3. Problem or Opportunity

Most people's financial data lives in too many places at once, bank apps that don't budget, spreadsheets that don't enforce structure, and mental notes that don't last. Without a single system, it's hard to see where money is actually going or whether savings goals are on track. ClearLedger solves this by consolidating accounts, transactions, budgets, and goals into one structured database.

4. Different From Competing Solutions

Tools like Mint, YNAB, and Personal Capital are powerful but come loaded with features most users don't need - automated bank syncing, credit score tracking, investment analytics. That complexity gets in the way. ClearLedger is intentionally narrow in scope: it does budgeting and transaction tracking, and it does them well. Because it's a custom-built relational database prototype, every data relationship and category is explicitly defined by the user, nothing is automated or opaque.

5. Prototype Scope

This prototype covers the core of what ClearLedger needs to be useful: user accounts, financial accounts (Checking, Savings, Credit, Investment), custom income and expense categories, transaction recording, monthly budgets, and savings goals. Seven report queries are included, covering monthly spending by category, income vs. expense summaries, budget vs. actual comparisons, savings goal progress, account activity, full transaction history, and a net worth view. Database maintenance scripts cover adding, editing, and deleting transactions, accounts, budgets, savings goals, and categories.

What's intentionally left out: automated bank integration, recurring transactions, budget alerts, financial forecasting, and multi-user access. These belong in the full application layer, not this prototype.

6. Database Design Overview

The database is named **clearledger** and is built on MySQL using InnoDB with utf8mb4 encoding. The schema has ten tables, six core tables and four lookup tables.

The core tables are user, account, category, budget, transaction, and savings_goal. The four lookup tables: account_type_lookup, transaction_type_lookup, category_type_lookup, and goal_status_lookup, replace free-form text fields with enforced, consistent values. All type and status columns in core tables reference these lookups via foreign keys.

Every table has an AUTO_INCREMENT primary key. Foreign key constraints enforce referential integrity across all relationships. A composite UNIQUE constraint on the budget table prevents a user from creating two budgets for the same category in the same month. A UNIQUE constraint on category prevents duplicate category names per user. The schema also includes one view, net_worth_summary_view, which aggregates each user's total income, total expenses, net cash flow, and total account balance.